

Quantum to Classical Transition

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Prerequisites:

Density-Matrix:

$$\langle n | \rho(t) | m \rangle \equiv \sum_{\alpha} p_{\alpha} \langle n | \Psi_{\alpha}(t) \rangle \langle \Psi_{\alpha}(t) | m \rangle$$

$$\rho = \sum_{\alpha} p_{\alpha} | \Psi_{\alpha}(t) \rangle \langle \Psi_{\alpha}(t) |$$

(Density-Matrix representation without choice of basis)

Where $\langle n |$ & $\langle m |$ are the basis and $\langle \Psi_{\alpha}(t) |$ is state of the system.

Property of Density-Matrix:

$$\begin{aligned}\langle O \rangle &= \sum_{\alpha} p_{\alpha} \langle \Psi_{\alpha} | O | \Psi_{\alpha} \rangle = \sum_{\alpha, m, n} p_{\alpha} \langle \Psi_{\alpha} | m \rangle \langle m | O | n \rangle \langle n | \Psi_{\alpha} \rangle \\ &= \sum_{m, n} \langle n | \rho | m \rangle \langle m | O | n \rangle = \text{tr}(\rho O)\end{aligned}$$

* Normalized : $\text{tr}(\rho) = 1$,

* Hermitian: $\rho^{\dagger} = \rho$

Reduced Density-Matrix:

Consider a system which is a composite of System A and System B.

Thus $H_{AB} = H_A \otimes H_B$, thus the corresponding density matrix ρ_{AB} .

Then the Density-Matrix of subsystem A is known as Reduced-Density Matrix is given as following:

$$\rho_A = \text{Tr}_B \rho_{AB}$$

Question/Motivation:

When does a quantum system transition to classical system?

DeCoherence

De-Coherence:

De-coherence is described as how entangling interaction with the environment influence the statistical results of future measurements on the system.

i.e Loss of quantum nature via entanglement with the environment

Double Slit Experiment:

THE DOUBLE SLIT EXPERIMENT

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The Final Composite State is : $|\Psi\rangle = \alpha|s_1\rangle|E_1\rangle + \beta|s_2\rangle|E_2\rangle$

Where $|s_1\rangle, |s_2\rangle$ are the states of the system S

$|s_1\rangle, |s_2\rangle$ state when particle when through slit 1, 2 respectively .

And $|E_1\rangle, |E_2\rangle$ are the state of the Environment E

(here the detector are the environment.)

$$\begin{aligned}
\rho_S &= \text{Tr}_E(\rho_{SE}) = \text{Tr}_E|\Psi\rangle\langle\Psi| \\
&= |\alpha|^2 |s_1\rangle\langle s_1| + |\beta|^2 |s_2\rangle\langle s_2| \\
&\quad + \alpha\beta^* |s_1\rangle\langle s_2| \langle E_2|E_1\rangle + \alpha^*\beta |s_2\rangle\langle s_1| \langle E_1|E_2\rangle.
\end{aligned}$$

Thus probability density of position x :

$$\begin{aligned}
p(x) &= \text{Tr}_S(\rho_S x) \\
&= |\alpha|^2 |\psi_1(x)|^2 + |\beta|^2 |\psi_2(x)|^2 \\
&\quad + 2 \text{Re} \{ \alpha\beta^* \psi_1(x) \psi_2^*(x) \langle E_2|E_1\rangle \} ,
\end{aligned}$$

Hence when $\langle E_1 | E_2 \rangle = 0$ (i.e perfectly distinguishable)

the interference term vanishes is now the system behave like classical particle

when $\langle E_1 | E_2 \rangle = 1$ (i.e perfectly In-distinguishable)

the interference term at Max and now the system behave like wave

Another way to think of Decoherence , is Indirect measurement/monitoring
Of the system by the Environment.

Motivation to alternative definition:

When $1 > \langle E_1 | E_2 \rangle > 0$ an interference experiment carried out on system S would reduce in visibility, representing a diminish in coherence between $|s_1\rangle$ and $|s_2\rangle$.

The Idea is more interaction with the environment eventually bring $\langle E_i(t) | E_j(t) \rangle = 0$

Decoherence models suggest $\langle E_i(t) | E_j(t) \rangle \propto e^{-t/\tau_d}$,

where τ_d is known as characteristic decoherence timescale

Mathematical formulation for Decoherence:

Consider at $t=0$ there are no correlation/entanglement between S and E

Thus $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0)$, $\rho_E(0) = \sum_i p_i |E_i\rangle \langle E_i|$ where $\sum_i p_i = 1$ and the states $|E_i\rangle$ forms orthonormal basis of Hilbert space E .

Time evolution operator $U = e^{-iHt}$, where H is total hamiltonian and $U^\dagger U = I$

Thus the Reduced density matrix of system S :

$$\begin{aligned} \rho_S(t) &= \text{Tr}_E \left\{ U(t) \left[\rho_S(0) \otimes \left(\sum_i p_i |E_i\rangle \langle E_i| \right) \right] U^\dagger(t) \right\} \\ &= \sum_{ij} p_i \langle E_j | U(t) | E_i \rangle \rho_S(0) \langle E_i | U^\dagger(t) | E_j \rangle. \quad (6) \end{aligned}$$

Introducing Kraus operators $E_{ij} \equiv \sqrt{p_i} \langle E_j | U | E_i \rangle$

$$\text{Thus } \rho_S(t) = \sum_{ij} E_{ij} \rho_S(0) E_{ij}^\dagger$$

Its customary to combine indices i and j into single index k .

$W_k \equiv \sqrt{p_i k} \langle E_{jk} | U(t) | E_{ik} \rangle$ which encodes information about initial state of E and the Dynamics of joint system SE .

thus :

$$\rho_S(t) = \sum_k W_k \rho_S(0) W_k^\dagger$$

Pointer States:

These are the least entangled states with the environment thus stay immune to de-coherence resulting in dynamic selection of preferred states.

For practical purpose $H = H_E + H_S + H_{int} \approx H_{int} = \sum_{\alpha} S_{\alpha} \otimes E_{\alpha}$

Pointer states $|s_i\rangle$ follow properties such as :

$$e^{-iH_{int}t} |s_i\rangle |E_0\rangle = \lambda_i |s_i\rangle e^{-iH_{int}t} |E_0\rangle \equiv |s_i\rangle |E_i(t)\rangle$$

hence one can construct $O_S = \sum_i o_i |s_i\rangle \langle s_i|$ which has to follow $[O_S, H_{int}] = 0$

If one could express $H_{int} = \sum_{\alpha} S_{\alpha} \otimes E_{\alpha}$, then $S_{\alpha} |s_i\rangle = \lambda_i |s_i\rangle$ is sufficient condition.

Example of a pointer states: Collisional Decoherence

- * Most of our daily observation depends on scattering of surrounding particles.
- * Scattering potentials almost always depend on distance x .
- * Thus the $[H_{int}, x] = 0$
- * Suggesting the pointer state of such a scenario is any form of $|x_i\rangle$.
- * We shall discuss more on this later next week.

Born-markov Master equation:

1) The Born approximation : The System environment coupling is sufficiently weak & the Environment is reasonably large such the changes in ρ_E is negligible. Thus $\rho_{SE} \approx \rho_S(t) \otimes \rho_E$.

2) Markov approximation: Memory effects of the environment are negligible i.e self correlation within environment created by the coupling to the system decay rapidly compared to the relaxation timescale of the open quantum system.

Assuming these conditions are true and $H_{int} = \sum_{\alpha} S_{\alpha} \otimes E_{\alpha}$, we get

$$\begin{aligned} \frac{d}{dt} \rho_S(t) = & -i [H_S, \rho_S(t)] \\ & - \sum_{\alpha} \{ [S_{\alpha}, B_{\alpha} \rho_S(t)] + [\rho_S(t) C_{\alpha}, S_{\alpha}] \}, \end{aligned} \quad (26)$$

where the system operators B_{α} and C_{α} are defined as

$$B_{\alpha} \equiv \int_0^{\infty} d\tau \sum_{\beta} c_{\alpha\beta}(\tau) S_{\beta}^{(I)}(-\tau), \quad (27a)$$

$$C_{\alpha} \equiv \int_0^{\infty} d\tau \sum_{\beta} c_{\beta\alpha}(-\tau) S_{\beta}^{(I)}(-\tau). \quad (27b)$$

Here $S_{\alpha}^{(I)}(-\tau)$ denotes the operator S_{α} in the interaction picture. In the following, we will simplify notation by

Appendix:

























